

6. 数据库约束

1. 本节目标

- 掌握约束的使用场景
- 掌握非空约束、默认约束、主键约束、唯一约束、外键约束、CHECK的创建

2. 什么是数据库约束

数据库约束是指对数据库表中的数据所施加的规则或条件，用于确保数据的准确性和可靠性。这些约束可以是基于数据类型、值范围、唯一性、非空等规则，以确保数据的正确性和相容性

3. 约束类型

类型	说明
NOT NULL非空约束	指定非空约束的列不能存储 NULL 值
DEFALUT 默认约束	当没有给列赋值时使用的默认值
UNIQUE 唯一约束	指定唯一约束的列每行数据必须有唯一的值
PRIMARY KEY 主键约束	NOT NULL 和 UNIQUE的结合，可以指定一个列或多个列，有助于防止数据重复和提高数据的查询性能
FOREIGN KEY 外键约束	外键约束是一种关系约束，用于定义两个表之间的关联关系，可以确保数据的完整性和一致性
CHECK 约束	用于限制列或数据在数据库表中的值，确保数据的准确性和可靠性

4. NOT NULL 非空约束

定义表时某列不允许为NULL时，可以为列添加非空约束。

- 比如创建一个学生表，学生名为NULL时，这条记录是不完整的

```
1 drop table if exists student;
2 create table student(
3     id bigint,
4     name varchar(20)
```

```

5 );
6
7 # 插入数据
8 insert into student values (1, null);
9
10 # 查询
11 select * from student;
12 +-----+-----+
13 | id   | name |
14 +-----+-----+
15 |    1 | NULL | # 班级名为NULL,没有意义
16 +-----+-----+
17 1 row in set (0.00 sec)

```

- 此时需要约束学生名的列不能为NULL

```

1 drop table if exists student;
2
3 # 为所有列添加非空约束
4 create table student (
5     id bigint,
6     name varchar(20) NOT NULL
7 );
8
9 # 由于name列有非空约束,插入NULL值时报错
10 insert into student values (1, null);
11 ERROR 1048 (23000): Column 'name' cannot be null
12
13 # 正常值可以成功插入
14 insert into student values (1, '张三');
15 Query OK, 1 row affected (0.00 sec)
16
17 select * from student;
18 +-----+-----+
19 | id   | name |
20 +-----+-----+
21 |    1 | 张三 |
22 +-----+-----+
23 1 row in set (0.00 sec)

```

- 查看表结构, NULL列为NO表示值不允许为NULL, YES表示值可以为NULL

```

1 desc student;

```

```

2 +-----+-----+-----+-----+-----+
3 | Field | Type          | Null | Key | Default | Extra |
4 +-----+-----+-----+-----+-----+
5 | id    | bigint        | YES  |     | NULL    |       |
6 | name  | varchar(10)   | NO   |     | NULL    |       |
7 +-----+-----+-----+-----+-----+
8 2 rows in set (0.01 sec)

```

5. DEFALUT 默认值约束

DEFAULT 约束用于向列中插入默认值，如果没有为列设置值，那么会将默认值设置到该列

- 重构学生表，新增年龄列

```

1 drop table student;
2
3 # 创建学生表，加入年龄列
4 create table student (
5     id bigint,
6     name varchar(20) not null,
7     age int
8 );
9

```

- 插入一条记录，没有设置默认约束时，不指定年龄的值时列为NULL

```

1 insert into student(id, name) values (1, '张三');
2
3 select * from student;
4 +-----+-----+-----+
5 | id  | name  | age  |
6 +-----+-----+-----+
7 | 1   | 张三  | NULL | # 年龄值为NULL
8 +-----+-----+-----+
9 1 row in set (0.00 sec)

```

- 重构学生表，为年龄的列加入默认约束

```

1 drop table student;
2
3 # 为年龄列加入默认约束

```

```

4 create table student (
5   id bigint,
6   name varchar(20) not null,
7   age int DEFAULT 18
8 );

```

- 插入一条记录，不指定年龄的值时列使用了默认值

```

1 select * from student;
2 +-----+-----+-----+
3 | id   | name  | age   |
4 +-----+-----+-----+
5 |    1 | 张三  | 18    |
6 +-----+-----+-----+
7 1 row in set (0.00 sec)

```

- 查看表结构，年龄列的默认值为18

```

1 desc student;
2 +-----+-----+-----+-----+-----+-----+
3 | Field | Type          | Null | Key | Default | Extra |
4 +-----+-----+-----+-----+-----+-----+
5 | id    | bigint        | YES  |     | NULL    |       |
6 | name  | varchar(20)   | NO   |     | NULL    |       |
7 | age   | int           | YES  |     | 18      |       |
8 +-----+-----+-----+-----+-----+-----+
9 4 rows in set (0.00 sec)

```

- 当手动明确指年龄列为NULL时列值为NULL

```

1 insert into student(id, name, age) values (2, '李四', NULL);
2 Query OK, 1 row affected (0.00 sec)
3
4 select * from student;
5 +-----+-----+-----+
6 | id   | name  | age   |
7 +-----+-----+-----+
8 |    1 | 张三  | 18    |
9 |    2 | 李四  | NULL  |
10 +-----+-----+-----+
11 2 rows in set (0.00 sec)

```

6. UNIQUE 唯一约束

指定了唯一约束的列，该列的值在所有记录中不能重复，比如一个人的身份证号，学生的学号等

- 重构学生表，新增学号列

```
1 drop table student;
2
3 # 学号列设置唯一约束
4 create table student (
5     id bigint,
6     name varchar(20) not null,
7     age int DEFAULT 18,
8     sno varchar(10)
9 );
```

- 不设置唯一约束时，学号可以重复

```
1 insert into student(id, name, sno) values (1, '张三', '100001');
2 Query OK, 1 row affected (0.00 sec)
3
4 insert into student(id, name, sno) values (2, '李四', '100001');
5 Query OK, 1 row affected (0.00 sec)
6
7 select * from student;
8 +-----+-----+-----+-----+
9 | id   | name  | age  | sno   |
10 +-----+-----+-----+-----+
11 |    1 | 张三  |   18 | 100001 |
12 |    2 | 李四  |   18 | 100001 |
13 +-----+-----+-----+-----+
14 2 rows in set (0.00 sec)
```

- 重构学生表，为学号列设置唯一约束

```
1 drop table student;
2
3 create table student (
4     id bigint,
5     name varchar(20) not null,
6     age int DEFAULT 18,
```

```
7 sno varchar(10) UNIQUE # 唯一约束
8 );
9
```

- 插入重复的学号时报错，唯一约束生效

```
1 insert into student(id, name, sno) values (1, '张三', '100001');
2 Query OK, 1 row affected (0.01 sec)
3
4 insert into student(id, name, sno) values (2, '李四', '100001');
5 ERROR 1062 (23000): Duplicate entry '100001' for key 'student.sno'
6
7 select * from student;
8 +-----+-----+-----+-----+
9 | id   | name  | age  | sno   |
10 +-----+-----+-----+-----+
11 | 1    | 张三  | 18   | 100001 |
12 +-----+-----+-----+-----+
13 1 row in set (0.00 sec)
```

- 查看表结构，Key列显示UNI表示唯一约束

```
1 desc student;
2 +-----+-----+-----+-----+-----+
3 | Field | Type          | Null | Key | Default | Extra |
4 +-----+-----+-----+-----+-----+
5 | id    | bigint        | YES  |     | NULL    |       |
6 | name  | varchar(20)   | NO   |     | NULL    |       |
7 | age   | int           | YES  |     | 18      |       |
8 | sno   | varchar(10)   | YES  | UNI | NULL    |       |
9 +-----+-----+-----+-----+-----+
10 4 rows in set (0.00 sec)
```

7. PRIMARY KEY 主键约束

主键约束唯一标识数据库表中的每条记录。

主键必须包含唯一的值，且不能包含 NULL 值。

每个表只能有一个主键，可以由单个列或多个列组成。

通常为每张表都指定一个主键，主键列建议使用BIGINT类型

- 重构学生表，为ID列添加非空和唯一约束

```
1 drop table student;
2
3 create table student (
4   id bigint not null unique,
5   name varchar(20) not null,
6   age int DEFAULT 18,
7   sno varchar(10) UNIQUE
8 );
```

- 查看表结构，添加了非空和唯一约束之后Key列显示PRI表示主键

```
1 desc student;
2 +-----+-----+-----+-----+-----+-----+
3 | Field | Type          | Null | Key | Default | Extra |
4 +-----+-----+-----+-----+-----+-----+
5 | id    | bigint        | NO   | PRI | NULL    |       | # 主键
6 | name  | varchar(20)   | NO   |     | NULL    |       |
7 | age   | int           | YES  |     | 18      |       |
8 | sno   | varchar(10)   | YES  | UNI | NULL    |       |
9 +-----+-----+-----+-----+-----+-----+
10 4 rows in set (0.00 sec)
```

- 当Id列的重复时会发生主键冲突

```
1 insert into student(id, name, sno) values (1, '张三', '100001');
2 Query OK, 1 row affected (0.01 sec)
3
4 insert into student(id, name, sno) values (1, '李四', '100001');
5 ERROR 1062 (23000): Duplicate entry '1' for key 'student.id'
```

- 通常把主键列设置为自动增长，让数据库维护主键值，

```
1 drop table student;
2
3 # 重构学生表
4 create table student (
5   id bigint PRIMARY KEY auto_increment, # 设置自增主键
6   name varchar(20) not null,
```

```

7 age int DEFAULT 18,
8 sno varchar(10) UNIQUE
9 );

```

- 插入数据时不设置主键列的值

```

1 # 主键列的值为NULL
2 insert into student(id, name, sno) values (NULL, '张三', '100001');
3 Query OK, 1 row affected (0.00 sec)
4
5 # 不指定主键
6 insert into student(name, sno) values ('李四', '100002');
7 Query OK, 1 row affected (0.01 sec)
8
9 # 主键列的值自动生成
10 select * from student;
11 +-----+-----+-----+-----+
12 | id | name | age | sno |
13 +-----+-----+-----+-----+
14 | 1 | 张三 | 18 | 100001 |
15 | 2 | 李四 | 18 | 100002 |
16 +-----+-----+-----+-----+
17 2 rows in set (0.00 sec)

```

- 查看表结构，Extra列显示auto_increment 表示自增

```

1 desc student;
2 +-----+-----+-----+-----+-----+-----+
3 | Field | Type | Null | Key | Default | Extra |
4 +-----+-----+-----+-----+-----+-----+
5 | id | bigint | NO | PRI | NULL | auto_increment |
6 | name | varchar(20) | NO | | NULL | |
7 | age | int | YES | | 18 | |
8 | sno | varchar(10) | YES | UNI | NULL | |
9 +-----+-----+-----+-----+-----+-----+
10 4 rows in set (0.00 sec)

```

- 如果某条记录写入失败，新生成的主键值将会作废

```

1 # 由于学号重复，产生了唯一冲突，导致插入失败，ID为3的主键值作废
2 insert into student(name, sno) values ('王五', '100002');

```



```

3 ERROR 1062 (23000): Duplicate entry '100002' for key 'student.sno'
4
5 # 修改学号后成功插入数据
6 insert into student(name, sno) values ('王五', '100003');
7 Query OK, 1 row affected (0.00 sec)
8
9 # 查询后发现新记录ID列的值为4
10 select * from student;
11 +-----+-----+-----+-----+
12 | id | name | age | sno |
13 +-----+-----+-----+-----+
14 | 1 | 张三 | 18 | 100001 |
15 | 2 | 王五 | 18 | 100002 |
16 | 4 | 王五 | 18 | 100003 | # 新插入的记录
17 +-----+-----+-----+-----+
18 3 rows in set (0.00 sec)

```

- 主键值可以不连续

```

1 # 手动指定一个值
2 insert into student(id, name, sno) values (100, '赵六', '100004');
3 Query OK, 1 row affected (0.01 sec)
4
5 select * from student;
6 +-----+-----+-----+-----+
7 | id | name | age | sno |
8 +-----+-----+-----+-----+
9 | 1 | 张三 | 18 | 100001 |
10 | 2 | 王五 | 18 | 100002 |
11 | 4 | 王五 | 18 | 100003 |
12 | 100 | 赵六 | 18 | 100004 |
13 +-----+-----+-----+-----+
14 4 rows in set (0.00 sec)
15
16 # 下一次自增从主键的最大值开始
17 insert into student(name, sno) values ('钱七', '100005');
18 Query OK, 1 row affected (0.01 sec)
19
20 select * from student;
21 +-----+-----+-----+-----+
22 | id | name | age | sno |
23 +-----+-----+-----+-----+
24 | 1 | 张三 | 18 | 100001 |
25 | 2 | 王五 | 18 | 100002 |
26 | 4 | 王五 | 18 | 100003 |

```

```

27 | 100 | 赵六 | 18 | 100004 |
28 | 101 | 钱七 | 18 | 100005 | # ID列的值是101
29 +-----+-----+-----+-----+
30 5 rows in set (0.00 sec)

```

- 主键或唯一键冲突时的更新操作，插入否则更新

```

1 # 语法
2 INSERT ... ON DUPLICATE KEY UPDATE column = value [, column = value] ...
3
4 # 插入ID为100, 学号为100100的学生记录时, 报主键冲突
5 insert into student(id, name, sno) values (100, '赵六', '100100');
6 ERROR 1062 (23000): Duplicate entry '100' for key 'student.PRIMARY'
7
8 # 可以使用以上语法, 如果插入时有冲突则更新当前列的值
9 insert into student(id, name, sno) values (100, '赵六', '100100')
10     -> ON DUPLICATE KEY UPDATE name = '赵六', sno = '100100';
11 Query OK, 2 rows affected (0.01 sec)
12 # 两行受影响, 表示删除了原来的记录, 又新写入了一条记录
13 # 与update student set name = '赵六', sno = '100100' where id = 100; 等效
14
15 select * from student;
16 +-----+-----+-----+-----+
17 | id | name | age | sno |
18 +-----+-----+-----+-----+
19 | 1 | 张三 | 18 | 100001 |
20 | 2 | 王五 | 18 | 100002 |
21 | 4 | 王五 | 18 | 100003 |
22 | 100 | 赵六 | 18 | 100100 | # 学号已修改
23 | 101 | 钱七 | 18 | 100005 |
24 +-----+-----+-----+-----+
25 5 rows in set (0.00 sec)

```

- 替换，如果存在冲突则替换，不存在冲突则插入

```

1 # 语法
2 REPLACE [INTO] table_name
3     [(column [, column] ...)]
4 VALUES
5     (value_list) [, (value_list)] ...
6
7 value_list: value, [, value] ...
8

```

```

9 # 写入或更新Id为101的记录
10 REPLACE into student(id, name, sno) values (101, '钱七', '100101');
11 Query OK, 2 rows affected (0.01 sec) # 受影响两行
12
13 # 原数据已更新
14 select * from student;
15 +-----+-----+-----+-----+
16 | id | name | age | sno |
17 +-----+-----+-----+-----+
18 | 1 | 张三 | 18 | 100001 |
19 | 2 | 王五 | 18 | 100002 |
20 | 4 | 王五 | 18 | 100003 |
21 | 100 | 赵六 | 18 | 100100 |
22 | 101 | 钱七 | 18 | 100101 |
23 +-----+-----+-----+-----+
24 5 rows in set (0.00 sec)
25
26 # 写入一条新数据
27 REPLACE into student(id, name, sno) values (102, '吴八', '100102');
28 Query OK, 1 row affected (0.01 sec) # 受影响一行
29
30 # 成功写入
31 select * from student;
32 +-----+-----+-----+-----+
33 | id | name | age | sno |
34 +-----+-----+-----+-----+
35 | 1 | 张三 | 18 | 100001 |
36 | 2 | 王五 | 18 | 100002 |
37 | 4 | 王五 | 18 | 100003 |
38 | 100 | 赵六 | 18 | 100100 |
39 | 101 | 钱七 | 18 | 100101 |
40 | 102 | 吴八 | 18 | 100102 |
41 +-----+-----+-----+-----+
42 6 rows in set (0.00 sec)

```

- 表中不能有多主键

```

1 drop table student;
2 # 重构学生表
3 create table student (
4     id bigint PRIMARY KEY auto_increment, # 定义主键
5     name varchar(20) PRIMARY KEY # 定义主键
6 );
7 ERROR 1068 (42000): Multiple primary key defined # 报错

```

- 复合主键：由多个列共同组成的主键，主键是否冲突以多个列的组成进行判定

```
1 drop table student;
2 # 重构学生表
3 create table student (
4   id bigint,
5   name varchar(20),
6   PRIMARY KEY (id, name) # 指定复合主键
7 );
8
9 # 插入数据
10 insert into student(id, name) values (1, '张三');
11 Query OK, 1 row affected (0.01 sec)
12
13 # 重复插入主键冲突，此时主键值由id和name两个列共同决定
14 insert into student(id, name) values (1, '张三');
15 ERROR 1062 (23000): Duplicate entry '1-张三' for key 'student.PRIMARY'
16
17 # 修改id值插入成功
18 insert into student(id, name) values (2, '张三');
19 Query OK, 1 row affected (0.00 sec)
20
21 select * from student;
22 +-----+
23 | id | name |
24 +-----+
25 | 1 | 张三 |
26 | 2 | 张三 |
27 +-----+
28 2 rows in set (0.00 sec)
```

8. FOREIGN KEY 外键约束

外键用于定义主表和从表之间的关系

外键约束主定义在从表的列上，主表关联的列必须是主键或唯一约束

当定义外键后，要求从表中的外键列数据必须在主表的主键或唯一列存在或为null。

学生表 (从表)

班级表 (主表)

两个表之间通过班级编号建立关系

编号	姓名	性别	班级编号
1	张三	男	1
2	李四	女	1
3	王五	男	2
4	赵六	男	3
5	钱七	女	3

编号	班级
1	java113
2	java78
3	java23
4	C++110
5	C++112

- 创建班级表(主表), 并初始化数据

```

1 drop table if exists class;
2 # 建表
3 create table class (
4   id bigint primary key auto_increment,
5   name varchar(20) not null
6 );
7
8 # 初始化数据
9 insert into class (name) values ('java01'), ('java02'), ('java03'), ('C++01'),
   ('C++02');
10 Records: 5  Duplicates: 0  Warnings: 0
11
12 select * from class;
13 +----+-----+
14 | id | name  |
15 +----+-----+
16 | 1  | java01 |
17 | 2  | java02 |
18 | 3  | java03 |
19 | 4  | C++01  |
20 | 5  | C++02  |
21 +----+-----+
22 5 rows in set (0.00 sec)

```

- 重构学生表(从表), 加入外键约束

```

1 # 语法
2 foreign key (id) references class(id)
3
4 drop table if exists student;
5 # 重构表
6 create table student(

```

```

7  id bigint PRIMARY KEY auto_increment,
8  name varchar(20) not null,
9  age int DEFAULT 18,
10 class_id bigint,
11 foreign key (class_id) references class(id) # 创建外键约束
12 );

```

- 查看表结构，Key列的值为**MUL**表示外键约束的列

```

1 desc student;
2 +-----+-----+-----+-----+-----+-----+
3 | Field      | Type          | Null | Key | Default | Extra          |
4 +-----+-----+-----+-----+-----+-----+
5 | id         | bigint        | NO   | PRI | NULL    | auto_increment |
6 | name       | varchar(20)   | NO   |     | NULL    |                |
7 | age        | int           | YES  |     | 18      |                |
8 | class_id   | bigint        | YES  | MUL | NULL    |                |
9 +-----+-----+-----+-----+-----+-----+
10 4 rows in set (0.00 sec)

```

- 正常插入数据

```

1 # 班级编号在主表中存在
2 insert into student(name, class_id) values ('张三', 1), ('李四', 2);
3 Query OK, 2 rows affected (0.01 sec)
4 Records: 2 Duplicates: 0 Warnings: 0
5
6 # 插入成功
7 select * from student;
8 +-----+-----+-----+-----+
9 | id | name  | age | class_id |
10 +-----+-----+-----+-----+
11 | 1  | 张三  | 18  | 1        |
12 | 2  | 李四  | 18  | 2        |
13 +-----+-----+-----+-----+
14 2 rows in set (0.00 sec)

```

- 插入一个班级号为100的学生，由于主表中没有这个班级，插入失败

```

1 # 班级编号在主表中不存在，提示外键约束限制导致插入失败
2 insert into student(name, class_id) values ('王五', 100);

```

```
3 ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint
  fails (`java01`.`student`, CONSTRAINT `student_ibfk_1` FOREIGN KEY
    (`class_id`) REFERENCES `class` (`id`))
```

- 插入班级Id为NULL的记录，可以成功，表示当前学生还没有分配班级

```
1 # 班级为NULL
2 insert into student(name, class_id) values ('王五', NULL);
3 Query OK, 1 row affected (0.00 sec)
4
5 select * from student;
6 +-----+-----+-----+-----+
7 | id | name  | age | class_id |
8 +-----+-----+-----+-----+
9 | 1 | 张三  | 18  | 1        |
10 | 2 | 李四  | 18  | 2        |
11 | 4 | 王五  | 18  | NULL     |
12 +-----+-----+-----+-----+
13 3 rows in set (0.00 sec)
```

- 删除主表某条记录时，从表中不能有对该记录的引用

```
1 # 删除从表中没有引用的记录，可以成功
2 delete from class where name = 'java03';
3 Query OK, 1 row affected (0.00 sec)
4
5 select * from class;
6 +-----+-----+
7 | id | name  |
8 +-----+-----+
9 | 1 | java01 |
10 | 2 | java02 |
11 | 4 | C++01  |
12 | 5 | C++02  |
13 +-----+-----+
14 4 rows in set (0.00 sec)
15
16 # 删除从表中引用的记录，失败
17 delete from class where name = 'java01';
18 ERROR 1451 (23000): Cannot delete or update a parent row: a foreign key
  constraint fails (`java01`.`student`, CONSTRAINT `student_ibfk_1` FOREIGN KEY
    (`class_id`) REFERENCES `class` (`id`))
```

- 删除主表时要先删除从表

```
1 # 从表存在是不能删除主表
2 drop table class;
3 ERROR 3730 (HY000): Cannot drop table 'class' referenced by a foreign key
  constraint 'student_ibfk_1' on table 'student'.
4
5 # 删除从表
6 drop table student;
7 Query OK, 0 rows affected (0.02 sec)
8
9 # 再删除主表, 成功
10 drop table class;
11 Query OK, 0 rows affected (0.01 sec)
```

9. CHECK 约束

可以应用于一个或多个列，用于限制列中可接受的数据值，从而确保数据的完整性和准确性。

在8.0.16开始全面支持CHECK约束，之前的版本会忽略CHECK的定义

- 重构学生表，有以下要求，年龄不能小于16岁，性别只能是男或女

```
1 drop table if exists student;
2
3 # 加入CHECK约束
4 create table student(
5     id bigint PRIMARY KEY auto_increment, # 设置自增主键
6     name varchar(20) not null,
7     age int DEFAULT 18,
8     gender char(1),
9     check (age >= 16),
10    check (gender = '男' or gender = '女')
11 );
12
13 # 正常插入数据
14 mysql> insert into student(name, age, gender) values ('张三', 17, '男'), ('李
  四', 19, '女');
15 Query OK, 2 rows affected (0.01 sec)
16 Records: 2 Duplicates: 0 Warnings: 0
17
18 mysql> select * from student;
19 +-----+-----+-----+-----+
20 | id | name  | age  | gender |
```



```

21 +-----+-----+-----+-----+
22 | 1 | 张三 | 17 | 男 |
23 | 2 | 李四 | 19 | 女 |
24 +-----+-----+-----+-----+
25 2 rows in set (0.00 sec)
26
27 # 插入年龄小于16岁的记录, 失败
28 mysql> insert into student(name, age, gender) values ('张三', 15, '男');
29 ERROR 3819 (HY000): Check constraint 'student_chk_1' is violated.
30
31 # 插入性别的值不是男或女的记录, 失败
32 mysql> insert into student(name, age, gender) values ('张三', 17, '1');
33 ERROR 3819 (HY000): Check constraint 'student_chk_2' is violated.

```

- 创建新表, c1的值不能为0, c2的值必须大于0, c3的值不小于c2

```

1 # 列与列之间也可以比较, 需要在单独一行中定义
2 create table t_check (
3   c1 int check(c1 <> 0),
4   c2 int check(c2 > 0),
5   c3 int,
6   check(c3 >= c2)
7 );
8
9 # 插入正常数据
10 mysql> insert into t_check values (-1, 3, 10);
11 Query OK, 1 row affected (0.01 sec)
12
13 mysql> select * from t_check;
14 +-----+-----+-----+
15 | c1 | c2 | c3 |
16 +-----+-----+-----+
17 | -1 | 3 | 10 |
18 +-----+-----+-----+
19 1 row in set (0.00 sec)
20
21 # c1 = 0时, 失败
22 mysql> insert into t_check values (0, 5, 6);
23 ERROR 3819 (HY000): Check constraint 't_check_chk_1' is violated.
24
25 # c2 <= 0时, 失败
26 mysql> insert into t_check values (2, -10, 10);
27 ERROR 3819 (HY000): Check constraint 't_check_chk_2' is violated.
28
29 # c3 < c2时, 失败

```

```
30 mysql> insert into t_check values (2, 10, 9);  
31 ERROR 3819 (HY000): Check constraint 't_check_chk_3' is violated.
```